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PLASMA ETCHING TOOLS AND SYSTEMS

Abstract

A method of processing a substrate that includes: loading the substrate into a plasma etch chamber, the substrate including a patterned hard mask layer and an underlying layer, the plasma etch chamber including: a chamber part having a surface including a refractory metal; and a first electrode; flowing a process gas into the plasma etch chamber; while flowing the process gas, applying a source power to the first electrode of the plasma etch chamber to generate a plasma in the plasma etch chamber; exposing the surface of the chamber part to the plasma to sputter the refractory metal from the surface of the chamber part; and exposing the substrate to the plasma to deposit the refractory metal onto a portion of the patterned hard mask layer and etch the underlying layer selectively to the patterned hard mask layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of U.S. application Ser. No. 17/832,897, filed on Jun. 6, 2022, which application is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present invention relates generally to a system and a method for processing a substrate, and, in particular embodiments, to plasma etching tools and systems.

BACKGROUND

[0003] Generally, a semiconductor device, such as an integrated circuit (IC) is fabricated by sequentially depositing and patterning layers of dielectric, conductive, and semiconductor materials over a substrate to form a network of electronic components and interconnect elements (e.g., transistors, resistors, capacitors, metal lines, contacts, and vias) integrated in a monolithic structure. Many of the processing steps used to form the constituent structures of semiconductor devices are performed using plasma processes.

[0004] The semiconductor industry has repeatedly reduced the minimum feature sizes in semiconductor devices to a few nanometers to increase the packing density of components. Accordingly, the semiconductor industry increasingly demands plasma processing technology to provide processes for patterning features with accuracy, precision, and profile control, often at atomic scale dimensions. Meeting this challenge, along with the uniformity and repeatability needed for high volume IC manufacturing, requires further innovations in plasma processing technology.

SUMMARY

[0005] In accordance with an embodiment of the present invention, a method of processing a substrate that includes: loading the substrate into a plasma etch chamber, the substrate including a patterned hard mask layer and an underlying layer, the plasma etch chamber including: a chamber part having a surface including a refractory metal; and a first electrode; flowing a process gas into the plasma etch chamber; while flowing the process gas, applying a source power to the first electrode of the plasma etch chamber to generate a plasma in the plasma etch chamber; exposing the surface of the chamber part to the plasma to sputter the refractory metal from the surface of the chamber part; and exposing the substrate to the plasma to deposit the refractory metal onto a portion of the patterned hard mask layer and etch the underlying layer selectively to the patterned hard mask layer.

[0006] In accordance with an embodiment of the present invention, a plasma etching system for a substrate including: an etch chamber; a substrate holder disposed in the etch chamber; a top electrode disposed in the etch chamber, the top electrode having a surface including a refractory metal; a bottom electrode connected to the substrate holder; a first radio frequency (RF) power source connected to the bottom electrode, the first RF power source being configured to generate a plasma in the etch chamber and to sputter the refractory metal from the surface of the top electrode; and a focus ring disposed on the substrate holder and configured to surround the substrate.

[0007] In accordance with an embodiment of the present invention, a plasma etching system for a substrate including: an etch chamber, where an upper wall inside the etch chamber including a refractory metal; a substrate holder disposed in the etch chamber; a top electrode including a helical coil disposed outside the etch chamber, the top electrode surrounding a upper portion of the etch chamber; a bottom electrode connected to the substrate holder; a radio frequency (RF) power source connected to the top electrode, the first RF power source being configured to generate a plasma in the etch chamber, the plasma being configured to sputter the metal from the upper wall; and a focus ring disposed on the substrate holder and configured to surround the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0009] FIG. **1** illustrates an example capacitively coupled plasma (CCP) processing system having a top electrode with a metal coating in accordance with various embodiments;

[0010] FIG. **2** illustrates an example inductively coupled plasma (ICP) processing system having a top plate with a metal coating in accordance with alternate embodiments;

[0011] FIG. **3** illustrates a cross-sectional view of a top electrode of a CCP processing system and a substrate positioned in an etch chamber in accordance with various embodiments, wherein a plasma in the etch chamber causes sputtering on the top electrode and reactive ion etching (RIE) over the substrate;

[0012] FIG. **4** illustrates a cross-sectional view of a top electrode of a CCP processing system and a substrate positioned in an etch chamber in accordance with alternate embodiments, wherein a plasma in the etch chamber causes deposition on the top electrode and reactive ion etching (RIE) over the substrate;

[0013] FIGS. **5**A and **5**B illustrate cross-sectional views of an example substrate during an example high aspect ratio (HAR) patterning process performed using a plasma processing system in accordance with various embodiments, wherein FIG. **5**A illustrates an incoming substrate with a patterned hard mask and a material layer, and FIG. **5**B illustrates the substrate after reactive ion etching (RIE) accompanied by deposition of a transition metal; and

[0014] FIG. **6** illustrates a process flow diagram of a reactive ion etching (RIE) process using a plasma processing system in accordance with one embodiment.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0015] This application relates to a system and method of processing a substrate through plasma etching and metal sputtering, which may be useful for fabrication processes for high-capacity three-dimensional (3D) memory devices, such as a 3D-NAND (or vertical-NAND), 3D-NOR, or dynamic random-access memory (DRAM) device. The fabrication of such devices may generally require forming conformal, high aspect ratio (HAR) features of a circuit element, for example, high aspect ratio contact (HARC) and high aspect ratio trench (HART). Features with aspect ratio (ratio of height of the feature to the width of the feature) higher than 20:1 are generally considered to be high aspect ratio features, and in some cases, fabricating a higher aspect ratio, such as 100:1, may be desired for advanced 3D semiconductor devices. This complexity is mainly caused by the limited mask height and etch selectivity of conventional mask materials such as amorphous carbon layer (ACL) and amorphous silicon. While new materials such as metals, metal nitride, metal carbide, and metal silicide may offer a better etch selectivity, the deposition of a thick film and patterning them suitable for a HAR etch process tend to be challenging. Therefore, a simple yet effective HAR process may be desired. Embodiments of the present application disclose systems and methods of fabricating HAR features by a plasma etch process that incorporates metal sputtering. Specifically, such plasma etching systems are characterized by at least one chamber part containing a metal element, for example, a refractory metal such as tungsten (W). The metal-containing chamber part may be a top electrode, a focus ring, a chamber wall, or other parts of the plasma processing system, and they may be configured to be sputtered in the presence of a plasma to provide the metal element to the plasma. The sputtered metal may then be deposited over a hard mask on a substrate to form a passivation layer, which may advantageously improve the etch selectivity.

[0016] In the following, example plasma etching systems with a metal-containing chamber part are first described, referring to FIGS. **1** and **2** for a capacitively coupled plasma (CCP) processing system and an inductively coupled plasma (ICP) processing system, respectively. Possible interactions of reactive species during plasma etching in a plasma etch chamber, with and without metal sputtering from the chamber part, are then described, referring to FIGS. **3** and **4**. Subsequently, FIGS. **5**A and **5**B illustrate steps of a plasma etching process for high aspect ratio (HAR) feature patterning utilizing metal sputtering for enhanced etch selectivity. An example process flow diagram is illustrated in FIG. **6**. All figures in this disclosure are drawn for illustration purposes only and not to scale, including the aspect ratios of features.

[0017] FIG. **1** illustrates an example capacitively coupled plasma (CCP) processing system **10** in accordance with various embodiments.

[0018] As illustrated in FIG. **1**, the CCP processing system **10** comprises a plasma etch chamber **110**, and a substrate **100** may be placed on a substrate holder **105**. In various embodiments, the substrate **100** may be a part of, or include, a semiconductor device, and may have undergone a number of steps of processing following, for example, a conventional process. The substrate **100**, accordingly, may comprise layers of semiconductors useful in various microelectronics. For example, the semiconductor structure may comprise the substrate **100** in which various device regions are formed.

[0019] In one or more embodiments, the substrate **100** may be a silicon wafer or a silicon-on-insulator (SOI) wafer. In certain embodiments, the substrate **100** may comprise a silicon germanium wafer, silicon carbide wafer, gallium arsenide wafer, gallium nitride wafer and other compound semiconductors. In other embodiments, the substrate **100** comprises heterogeneous layers such as silicon germanium on silicon, gallium nitride on silicon, silicon carbon on silicon, as well as layers of silicon on a silicon or SOI substrate. In various embodiments, the substrate **100** is patterned or embedded in other components of the semiconductor device.

[0020] One or more process gases may be introduced into the plasma etch chamber **110** by a gas delivery system **115**. The gas delivery system **115** may comprise multiple gas flow controllers to control the flow of multiple gases into the plasma etch chamber **110**. In some embodiments, optional center/edge splitters may be used to independently adjust the gas flow rates at the center and edge of the substrate **100**. Further, in one embodiment, the gas delivery system **115** may have a special showerhead configuration positioned at the top of the plasma etch chamber **110**. For example, the gas delivery system **115** may be integrated with a top electrode **150**, having a showerhead configuration on the top electrode **150**, covering the entirety of the substrate **100**, including a plurality of appropriately spaced gas inlets. Alternatively, gas may be introduced through dedicated gas inlets or any other suitable configuration. The plasma etch chamber **110** may further be equipped with one or more sensors such as pressure monitors, gas flow monitors, and/or gas species density monitors. The sensors may be integrated as a part of the gas delivery system **115** in various embodiments.

[0021] In FIG. **1**, the plasma etch chamber **110** is a vacuum chamber and may be evacuated using one or more vacuum pumps **135**, such as a single-stage pumping system or a multistage pumping system (e.g., a mechanical roughing pump combined with one or more turbomolecular pumps). To promote even gas flow during plasma processing, gas may be removed from more than one gas outlet or location in the plasma etch chamber **110** (e.g., on opposite sides of the substrate **100**).

[0022] In various embodiments, the substrate holder **105** may be integrated with, or a part of, a chuck (e.g., a circular electrostatic chuck (ESC)) positioned near the bottom of the plasma etch chamber **110**, and connected to a bottom electrode **120**. The surface of the chuck or the substrate holder **105** may be coated with a conductive material (e.g., a carbon-based or metal-nitride based coating). The substrate **100** may be optionally maintained at a desired temperature using a temperature sensor and a heating element connected to a first temperature controller **140**. In certain embodiments, the temperature sensor may comprise a thermocouple, a resistance temperature detector (RTD), a thermistor, or a semiconductor based integrated circuit. The heating element may, for example, comprise a resistive heater in one embodiment. In addition, there may be a cooling element such as a liquid cooling system coupled to the first temperature controller **140**. The bottom electrode **120** may be connected to one or more RF power sources **130** to generate a plasma **160** in the plasma etch chamber **110**. As illustrated in FIG. **1**, more than one RF power source may be used, for example, to provide a high-frequency RF power (HF) and a low-frequency RF power (LF) at the same time. In various embodiments, the HF may be used for plasma and radical generation, the LF may be used for ion acceleration in a sheath of the plasma **160** over the substrate **100**, which enables plasma etching on the substrate **100**. In certain embodiments for a CCP processing system, the frequency of the HF may range from 27 MHz to 150 MHz, and that of the LF may range from 400 kHz to 13 MHz. The RF power sources **130** may be used to supply continuous wave (CW) or pulsed RF power to sustain the plasma **160**. The plasma **160**, shown between a top electrode **150** and the bottom electrode **120**, exemplifies direct plasma generated close to the substrate **100** in the plasma etch chamber **110**.

[0023] In various embodiments, an RF pulsing at a kHz range may be used to power the plasma **160**. Using the RF pulsing may help generate high energetic ions (>keV) in the plasma **160** for the plasma etch process, while reducing the charging effect. The charging effect during a process is a phenomenon where electrons build charge on insulating materials, creating a local electric field that may steer positive ions to the sidewalls and cause a lateral etching. Therefore, fine-tuning the power conditions of the plasma etch process may also be important to minimize the widening of the critical dimension (CD) of the high aspect ratio (HAR) feature. In certain embodiments, a moderate duty ratio between 10% and 100% may be used. In one embodiment, a bias power of 18 kW may be pulsed at a frequency of 5 kHz with a duty ratio of 60%.

[0024] Further illustrated in FIG. **1**, the top electrode **150** may be a conductive circular plate inside the plasma etch chamber **110** near the top. In various embodiments, the top electrode **150** may be connected to a direct current (DC) voltage source **165** of the CCP processing system **10**. Combined with the RF power from the RF power sources **130**, the DC voltage is used to generate a DC superimposed RF plasma in the plasma etch chamber **110**. In FIG. **1**, the DC voltage may be supplied to the top electrode **150**. In another embodiment, the DC voltage may be supplied to the bottom electrode **120**. In various embodiments, the DC voltage may advantageously be adjusted to tune the degree of metal sputtering and thereby the concentration of the metal elements in the plasma **160**.

[0025] The DC voltage supplied by the DC voltage source **165** can range from positive to negative. A negative DC voltage at the top electrode **150** may advantageously adjust (e.g., increase) the average ion energy of species of the plasma **160**. In various embodiments, the DC voltage V.sub.DC coupled to the top electrode **150** may be in the range of 0 V to about 3000 V. In one embodiment, the DC voltage V.sub.DC coupled to the top electrode **150** may be about −200 V. In further embodiments, instead of the DC voltage source **165**, another RF power source may be used and configured to provide RF power to the top electrode **150**. In one or more embodiments, the frequency for the RF power to the top electrode **150** may range from 400 kHz to 13 MHz.

[0026] In various embodiments, the CCP processing system **10** is particularly characterized by the top electrode **150** comprising a metal, for example, with a metal-containing coating **152**. During a plasma processing such as reactive ion etching (RIE) using the CCP processing system **10**, the metal of the metal-containing coating **152** may advantageously be sputtered by ion bombardment, resulting in the plasma **160** containing the metal. This metal sputtered into the plasma **160** may then be deposited on a hard mask present on the substrate **100** to form a passivation layer, which may enhance the etch selectivity, as further described referring to FIGS. **3-5**B. In various embodiments, the metal may be a transition metal. In certain embodiments, the metal may be tungsten (W). Examples of the metals useful for the metal-containing coating **152** further include titanium (Ti) and tantalum (Ta), but other metals may also be used. Generally, elements that exhibit etch resistance, when deposited from the plasma **160**, better than the etch mask used in the process, are useful and preferred. The metal-containing coating **152** may be in pure metal form in certain embodiments. Still, in other embodiments, it may be metal carbide (e.g., WC), metal nitride (e.g., WN), metal silicide (WSi.sub.x), or other metal compounds. Further, the metal-containing coating **152** may also comprise a thin layer of an oxide on the surface.

[0027] The top electrode **150** may, in one or more embodiments, be connected to a second temperature controller **155** configured to control the temperature of the top electrode **150** and the metal-containing coating **152**. The second temperature controller **155** may further comprise or be coupled to a temperature sensor and a heating element. In certain embodiments, the temperature sensor may comprise a thermocouple, a resistance temperature detector (RTD), a thermistor, or a semiconductor based integrated circuit. The heating element may, for example, comprise a resistive heater in one embodiment. In addition, there may be a cooling element such as a liquid cooling system coupled to the second temperature controller **155**. Since the metal sputtering may depend on the temperature of the target, controlling the temperature of the top electrode **150** may be useful in adjusting the degree of metal sputtering. For example, increasing the temperature of the metal-containing coating **152** may enhance the metal sputtering by the plasma **160**.

[0028] Although the metal-containing coating **152** covers the bottom surface of the top electrode **150** in FIG. **1**, in alternate embodiments, the metal may be incorporated in a chamber part of the plasma etch chamber **110** in any reasonable fashion as long as the metal sputtering may be enabled. Accordingly, the metal may be incorporated by, for example, the top electrode **150** entirely made of, plated with, brazed with, or deposited with the metal, metal carbide, metal nitride, metal silicide, or other metal compounds.

[0029] In various embodiments, the CCP processing system **10** may further comprise a focus ring **154** positioned over the bottom electrode **120** to surround the substrate **100**. The focus ring **154** may advantageously maintain and extend the uniformity of the plasma **160** to achieve process consistency at the edge of the substrate **100**. In various embodiments, the focus ring **154** may have a width of a few cm. In various embodiments, there may be a gap for mechanical clearance between the circumference of the substrate **100** and the focus ring **154**. In certain embodiments, the gap may be hundreds of microns to a few mm. In various embodiments, the focus ring **154** may comprise a dielectric material with a desired dielectric constant. In certain embodiments, the focus ring **154** may comprise silicon. Some examples of silicon-based focus rings may comprise silicon, silicon oxide, doped silicon (e.g., boron-doped, nitrogen-doped, and phosphorus-doped), or silicon carbide. Alternatively, in some embodiments, the focus ring may comprise a carbon-based material.

[0030] In certain embodiments, similar to the top electrode **150** as described above, the focus ring **154** may also comprise a focus ring metal-containing coating **156** on the surface. The metal of the focus ring metal-containing coating **156** may be utilized as an additional source of metal for the metal sputtering during a plasma process. The metal used in the focus ring metal-containing coating **156** may be tungsten (W), titanium (Ti), tantalum (Ta), or other metals. In one embodiment, the metal of the focus ring metal-containing coating **156** and the metal of the metal-containing coating **152** may be the same. Still, in another embodiment, they may be different. The metal of the focus ring metal-containing coating **156** may be in pure metal form in certain embodiments. Still, in other embodiments, it may be metal carbide (e.g., WC), metal nitride (e.g., WN), metal silicide (WSi.sub.x), or other metal compounds. In one or more embodiments, the focus ring **154**, instead of having the focus ring metal-containing coating **156**, may be entirely made of, plated with, brazed with, or deposited with the metal, metal carbide, metal nitride, metal silicide, or other metal compounds.

[0031] As illustrated in FIG. **1**, the focus ring **154** may be connected to an RF power source **170** configured to apply an RF bias to the focus ring **154** in one embodiment. In another embodiment, a

DC voltage source may be used instead of an RF power source. Applying a bias to the focus ring **154** may advantageously improve the uniformity of the plasma **160** in the plasma etch chamber **110** and also adjust the degree of the metal sputtering, particularly from the surface of the focus ring metal-containing coating **156** and/or the focus ring **154**. The first temperature controller **140** may also be configured to control the temperature of the focus ring **154** in certain embodiments. Controlling the temperature of the focus ring metal-containing coating **156**, similar to the top electrode **150**, may be useful in adjusting the degree of metal sputtering.

[0032] Although the incorporation of the metal to the top electrode **150** and the focus ring **154** is described above, any suitable chamber parts, including but not limited to a chamber wall, may be fabricated to include the metal on or near their surface and utilized as a metal source for metal sputtering.

[0033] FIG. **2** illustrates an example inductively coupled plasma (ICP) processing system **20** in accordance with alternate embodiments. For illustration purposes, some parts of the ICP processing system **20** that may be in common with the CCP processing system **10** illustrated in FIG. **1** are omitted (e.g., the first temperature controller **140** and the RF power source **170**) and not repeated below.

[0034] In FIG. **2**, the ICP processing system **20** comprises a plasma etch chamber **210** configured to sustain a plasma **260** directly above a substrate **200** loaded onto a substrate holder **205**. A focus ring **254** may be positioned to surround the substrate **200**. A process gas may be introduced to the plasma etch chamber **210** through a gas inlet connected to a gas flow control system **215** and may be pumped out of the plasma etch chamber **210** through a gas outlet connected to a vacuum pump **235**. The gas flow control system **215** may comprise various components such as high-pressure gas canisters, valves (e.g., throttle valves), pressure sensors, gas flow sensors, vacuum pumps, pipes, and electronically programmable controllers. A bottom electrode **220** may be connected to an RF bias power source **230**. For ICP configuration, a top electrode **250** may be a conductive helical coil electrode located outside the plasma etch chamber **210**, coiled around a dielectric sidewall **216**. The top electrode **250** may be connected to an RF source **265**. As further illustrated in FIG. **2**, the gas inlet is an opening in a top plate **212** and the gas outlet is an opening in a bottom plate **214**. The top plate **212** and bottom plate **214** may be conductive and electrically connected to the system ground (a reference potential).

[0035] In various embodiments, the ICP processing system **20** is particularly characterized by the top plate **212** comprising a metal, for example, with a metal-containing coating **252** inside the plasma etch chamber **210**. Unlike the prior embodiments of the CCP processing system **10**, the top electrode **250** is located outside the plasma etch chamber **210**. Accordingly, the metal-containing chamber part for metal sputtering may be, for example, the top plate **212** rather than the top electrode **250**. The metal-containing coating **252** may function as a metal source for the metal sputtering during plasma processing, such as reactive ion etching (RIE), using the ICP processing system **20**. In various embodiments, the metal may be tungsten (W), titanium (Ti), tantalum (Ta), or other metals. The metal-containing coating **252** may be in pure metal form in certain embodiments. Still, in other embodiments, it may be metal carbide (e.g., WC), metal nitride (e.g., WN), metal silicide (WSi.sub.x), or other metal compounds.

[0036] Similar to the prior embodiments, the focus ring **254** may also comprise a focus ring metal-containing coating **256**, which may serve an additional metal source for metal sputtering.

[0037] Although not specifically illustrated in FIG. **2**, the ICP processing system **20** may further comprise any additional components useful for the plasma processing, such as temperature controllers for the substrate **200**, the bottom electrode **220**, and/or the focus ring **254**, and an RF power source for the focus ring **254**. In one or more embodiments, the ICP processing system **20** may further comprise a direct current (DC) voltage source or a RF power source configured to provide a DC voltage or a RF power to a chamber part (e.g., the top plate **212**) that is isolated from the primary RF system to generate the plasma **260**. Such additional voltage or power sources may advantageously tune the degree of metal sputtering and thereby the concentration of the metal elements in the plasma **260**

[0038] The configurations of the plasma etching systems (e.g., the CCP processing system **10** and the ICP processing system **20**) described above are for example only. In alternative embodiments, various alternative configurations may be used for a plasma processing system that incorporates a metal-containing chamber part for metal sputtering. In an alternate embodiment for an ICP processing system, a conductive helical coil electrode (e.g., the top electrode **250** in FIG. **2**) may be located above the top plate **212** rather than the sidewall **216**. In this configuration, the top plate **212** may be made of a dielectric material to allow the electromagnetic field to be coupled to the plasma **260** in the plasma etch chamber **210**. Accordingly, unlike FIG. **2**, the focus ring metal-containing coating **256** may be applied to the sidewall **216** rather than the top plate **212**. In another example, the plasma processing system may be a resonator, such as a helical resonator. Further, microwave plasma (MW) or other suitable systems may be used. Pulsed RF power sources and pulsed DC voltage sources may also be used in some embodiments (as opposed to continuous wave RF power sources). In various embodiments, the RF power, chamber pressure, substrate temperature, gas flow rates and other plasma process parameters may be selected in accordance with the respective process recipe.

[0039] In various embodiments, a plasma etching process such as reactive ion etching (RIE) may comprise controlling the plasma condition to enable/disable the metal sputtering from the metal-containing chamber part, as further described referring to FIGS. **3** and **4**.

[0040] FIG. **3** illustrates a cross sectional view of a top electrode **150** of a CCP processing system and a substrate **100** positioned in a etch chamber in accordance with various embodiments, wherein a plasma **160** in the etch chamber causes sputtering on the top electrode **150** and reactive ion etching (RIE) over the substrate **100**.

[0041] In FIG. **3**, a material layer **310** may be formed over the substrate **100**. In various embodiments, the material layer **310** is a target layer that is to be patterned into one or more high aspect ratio (HAR) features. In certain embodiments, the HAR feature being etched into the material layer **310** may be a contact hole, slit, or other suitable structures comprising a recess. In one embodiment, the material layer **310** may be a silicon oxide layer. The material layer **310** may be deposited using an appropriate technique such as vapor deposition including chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), as well as other plasma processes such as plasma enhanced CVD (PECVD) and other processes. In one embodiment, the material layer **310** has a thickness between 0.1 μm and 100 μm.

[0042] Still referring to FIG. **3**, a patterned layer **320** is formed over the material layer **310**. In various embodiments, the patterned layer **320** may comprise any material useful to enable patterning of the material layer **310** by subsequent patterning processes. In various embodiments, the patterned layer **320** may comprise a photoresist, an organic dielectric layer (ODL), or an amorphous carbon layer (ACL). The patterned layer **320** may comprise a hard mask, including but not necessarily limited to amorphous silicon, silicon oxide, silicon nitride, or a metal-based hard mask. In one or more embodiments, the patterned layer **320** may be a layer stack comprising multiple layers, for example, a tri-layer stack commonly used for a photolithographic process. The patterned layer **320** may be formed by first depositing a hard mask layer using, for example, an appropriate spin-coating technique or a vapor deposition technique such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), as well as other plasma processes such as plasma enhanced CVD (PECVD) and other processes. The deposited hard mask layer may then be patterned using a lithography process and an anisotropic etch process. The relative thicknesses of the patterned layer **320** and the material layer **310** may have any suitable relationship. For example, the patterned layer **320** may be thicker than the material layer **310**, thinner than the material layer **310**, or the same thickness as the material layer **310**. In one embodiment, the patterned layer **320** has a thickness between 0.1 μm and 10 μm.

[0043] The patterned layer **320** and/or the material layer **310** may be collectively considered as a part of the substrate **100**. Further, the substrate **100** may also comprise other layers. For example, to pattern the layer, a tri-layer structure comprising a photoresist layer, SiON layer, and optical planarization layer (OPL) may be present.

[0044] Fabricating the HAR feature in the material layer **310** may be performed by a plasma etch process using a combination of process gases to generate the plasma **160**. In various embodiments, the process gas may comprise any reasonable gas that may provide an etchant for the plasma etch process, for example, a halogen. In certain embodiments, the process gas may comprise fluorocarbon or hydrofluorocarbon. Examples of such process gases include tetrafluoromethane (CF_4), trifluoromethane (CHF_3), difluoromethane (CH_2F_2), octafluoropropane (C_3F_8), hexafluoropropylene (C_3F_6), perfluorobutane (C_4F_{10}), octafluorocyclobutane (C_4F_8), octafluoro-2-butene (C_4F_8), perflenapent (C_5F_{12}), hexafluorobutadiene (C_4F_6), hexafluoro-2-butyne (C_4F_6), and hexafluorocyclobutene (C_4F_6). In certain embodiments, other gases such as a noble gas and/or a balancing agent may also be added. For example, in certain embodiments, argon (Ar) and dioxygen (O_2) may be included as the noble gas and the balancing agent, respectively.

[0045] Accordingly, in certain embodiments, the plasma **160** may comprise positively charged species **302** (e.g., Ar^+), carbon species **306**, and fluorine species **308** as illustrated in FIG. **3**. Here, the sputtering of the metal from the metal-containing coating **152** of the top electrode **150** may be enabled when the positively charged species **302** is provided with sufficient ion bombardment energy. The positively charged species **302** impinges on the metal-containing coating **152** as indicated by an arrow in FIG. **3** and metal species **304** may be sputtered into the plasma.

[0046] Various process parameters may be utilized to enable and control the metal sputtering. For example, increasing the DC voltage applied to the top electrode **150** may increase the ion bombardment energy of the positively charged species **302** and thereby their sputtering ability. In addition, the degree of metal sputtering may also depend on the temperature of the metal-containing coating **152**, where a higher temperature leads to a greater amount of sputtering. A precise, local temperature control may therefore be utilized to control the metal sputtering. Other parameters such as process time, gas composition (e.g., inert gas concentration), chamber pressure, RF source power, and RF bias power may also impact the metal sputtering, and thus may be selected accordingly in a process recipe.

[0047] Still referring to FIG. **3**, with the plasma **160** enabling reactive ion etching (RIE), a recess is formed in the material layer **310** according to the pattern of the patterned layer **320**. In various embodiments, the fluorine species **308** may function as the primary etchant. During this etching, a passivation layer **330** may be formed on the surface of the patterned layer **320**, advantageously preventing the hard mask to be etched and thereby improving the etch selectivity. In various embodiments, the passivation layer **330** may be formed through deposition of the metal species **304** and the carbon species **306**, but in other embodiments, other species may be involved. In one or more embodiments, the passivation layer **330** may comprise a metal carbide, for example, tungsten carbide (WC), which exhibits high mechanical and chemical stability. In various embodiments, the passivation layer **330** may comprise carbon originating from the carbon species **306**, the hard mask material, or both. If the hard mask is a silicon-based material, only the carbon species **306** may be the source for the carbon in the passivation layer **330**.

[0048] FIG. **4** illustrates a cross-sectional view of a top electrode **150** of a CCP processing system and a substrate **100** positioned in an etch chamber in accordance with alternate embodiments, wherein a plasma **160** in the etch chamber causes deposition on the top electrode **150** and reactive ion etching (RIE) over the substrate **100**. The structure of the substrate **100** and species of the plasma **160** are the same as those illustrated in FIG. **3** and will not be repeated.

[0049] In FIG. **4**, the condition of the plasma **160** may be changed from that of FIG. **3** to disable the metal sputtering while maintaining the etching ability. As illustrated in FIG. **4**, the metal sputtering by ion bombardment may be disabled by the formation of a deposition layer **410**, for example, comprising carbon. In various embodiments, the deposition layer **410** may comprise carbon materials formed through the deposition of the carbon species **306** of the plasma **160**. The deposition layer **410** may passivate and protect the metal-containing coating **152** from being sputtered by ions such as the positively charged species **302** (e.g., Ar.sup.+). In one or more embodiments, this no-sputtering condition may be realized by turning off or decreasing the DC voltage V.sub.dc applied to the top electrode **150**, which reduces the ion bombardment energy of species of the plasma **160**. In addition, other process parameters may be controlled to disable the metal sputtering. In one embodiment, the metal-containing coating **152** may be cooled. In certain embodiments, the process conditions may disable the metal sputtering without the formation of the deposition layer **410**.

[0050] As described above referring to FIGS. **3** and **4**, the plasma process conditions used for etching the material layer **310** may or may not enable the metal sputtering from the metal-containing chamber part (e.g., the top electrode **150** having the metal-containing coating **152**), primarily depending on the ion bombardment energy provided to the plasma **160**. Generally, higher ion bombardment energy may be desired for a faster etch rate and to enable metal sputtering. Still, it may impair the etch selectivity and the formation of the passivation layer **330**. Accordingly, addressing this trade-off, the plasma etching process in various embodiments may be a multi-step process employing multiple plasma conditions, for example, a cyclic process repeating sputtering and non-sputtering conditions. Therefore, embodiments may advantageously allow balancing the etching rate and the etch selectivity.

[0051] Further, in various embodiments, the process conditions and process recipe may be selected to achieve a desired thickness of the passivation layer **330**. While the presence of the passivation layer **330** is beneficial in improving the etch selectivity to the hard mask, excessive deposition of the passivation layer **330** may cause undesired critical dimension (CD) shrinkage and/or clogging issues. Accordingly, process parameters such as process time, gas composition (e.g., inert gas concentration), chamber pressure, RF source power, and RF bias power may be selected to balance the degree of metal deposition to form the passivation layer **330** as well as the metal sputtering.

[0052] FIGS. **5**A and **5**B illustrate cross-sectional views of an example substrate during an example high aspect ratio (HAR) patterning process using a plasma processing system in accordance with various embodiments.

[0053] FIG. **5**A illustrates a cross-sectional view of an incoming substrate **100** with a patterned layer **320** and a material layer **310**, similar to those illustrated in FIGS. **3** and **4**, and thus will not be repeated in detail. The material layer **310** is to be patterned to form HAR features that may be useful in high-capacity three-dimensional (3D) memory devices, such as a 3D-NAND (or vertical-NAND), 3D-NOR, or dynamic random-access memory (DRAM) device. These devices typically require forming conformal, high aspect ratio contact holes (HARC) or trenches (HART). In certain embodiments, the material layer **310** is a layer stack comprising multiple layers, for example, alternating oxide layers and nitride layers. As illustrated in FIG. **5**, the patterned layer **320** is characterized by having recesses **510**.

[0054] FIG. **5**B illustrates a cross-sectional view of the substrate **100** after reactive ion etching (RIE).

[0055] The plasma etch process may be a single-step process or a multi-step process, including a cyclic process, and may be fluorocarbon or hydrofluorocarbon-based as described above. In FIG. **5**B, the HAR feature is being formed by extending the recesses **510** into the material layer **310** by the plasma etch process. As described, referring to FIGS. **3** and **4**, since the plasma processing system is capable of metal sputtering from the metal-containing chamber part, a highly anisotropic plasma etch may be achieved. This is due to the metal sputtering and the metal deposition on the hard mask to form a passivation layer **330** comprising the metal (e.g., metal carbide). In certain embodiments, some other polymeric deposition (e.g., sidewall deposition on the material layer **310**)

may also occur.
[0056] The passivation layer **330** may comprise the metal from the sputtered metal, and carbon from the species in the plasma, the hard mask, or both. In various embodiments, the passivation layer **330** may comprise metals, metal nitride, metal carbide, or metal silicide. In one embodiment, the passivation layer **330** may comprise metal carbide (e.g., WC). The use of these metal-containing materials specifically for the passivation layer **330** over a conventional hard mask material may be beneficial in efficient HAR patterning processes. While these metal-containing materials may be potentially used as a hard mask itself (e.g., the patterned layer **320**) to offer excellent etch selectivity, depositing a sufficiently thick film and patterning these metal-containing materials as a hard mask may be challenging. In this approach, along with new materials, completely new techniques for deposition and for patterning may have to be developed. In contrast, various embodiments of this disclosure integrate a thin film of the metal-containing material (e.g., as the passivation layer **330**) with conventional hard mask materials (e.g., amorphous carbon and/or amorphous silicon), which are significantly easier to process and pattern as a hard mask. As a result, the etch resistance of the conventional hard mask materials may be substantially improved with minimal additional steps. The improved etch selectivity to the hard mask during the plasma etch process can thus reduce the consumption of the hard mask. Further, providing a metal-containing chamber part to the plasma processing system, the methods may not require including any metal element in the process gas. In various embodiments, the conventional process gas, such as fluorocarbon, for HAR patterning processes may be utilized with little to no modification.
[0057] Further illustrated in FIG. **5**B, the recesses **510** may reach to the top surface of the substrate **100**. The plasma etch process in accordance with various embodiments may provide a good selectivity to the material of the substrate **100** (e.g., silicon) in addition to the hard mask. Accordingly, the formation of the recesses **125** may advantageously stop at the top surface of the substrate **100**. Once the plasma etch process to form the HAR feature is completed, appropriate subsequent fabrication steps may be followed accordingly to, for example, a conventional process recipe. Such steps may include, but are not necessarily limited to, a removal of the remaining hard mask, a metallization, a staircase etch to form a staircase structure in the material layer **310** in the case of fabricating a 3D NAND device.
[0058] FIG. **6** illustrates a process flow diagram of a reactive ion etching (RIE) process in accordance with one embodiment.
[0059] In FIG. **6**, a process flow **60** starts with providing a substrate in a plasma etch chamber (block **610**, FIG. **5**A), where the substrate comprises a patterned hard mask layer and an underlying layer and the plasma etch chamber comprises a top electrode having a surface comprising a transition metal (e.g., W). Next, a process gas for the RIE process may be flowed into the plasma etch chamber (block **620**). While flowing the process gas, a source power is then applied to a bottom electrode of the plasma etch chamber to generate a plasma in the plasma etch chamber (block **630**). Subsequently, the surface of the top electrode may be exposed to the plasma to sputter the transition metal from the surface of the top electrode (block **640**). Simultaneously or subsequently, the substrate may be exposed to plasma to deposit the transition metal onto a portion of the patterned hard mask layer and etch the underlying layer selectively to the patterned hard mask layer (block **650**, FIG. **5**B).
[0060] Example embodiments of the invention are summarized here. Other embodiments can also be understood from the entirety of the specification as well as the claims filed herein.
[0061] Example 1. A method of processing a substrate that includes: loading the substrate into a plasma etch chamber, the substrate including a patterned hard mask layer and an underlying layer, the plasma etch chamber including: a chamber part having a surface including a refractory metal; and a first electrode; flowing a process gas into the plasma etch chamber; while flowing the process gas, applying a source power to the first electrode of the plasma etch chamber to generate a plasma in the plasma etch chamber; exposing the surface of the chamber part to the plasma to sputter the

refractory metal from the surface of the chamber part; and exposing the substrate to the plasma to deposit the refractory metal onto a portion of the patterned hard mask layer and etch the underlying layer selectively to the patterned hard mask layer.

[0062] Example 2. The method of example 1, where the refractory metal is tungsten, molybdenum, niobium, tantalum, or ruthenium.

[0063] Example 3. The method of one of examples 1 or 2, where the plasma is an inductively coupled plasma (ICP), and where the chamber part is a top plate disposed at an upper wall of the plasma etch chamber.

[0064] Example 4. The method of one of examples 1 or 2, where the plasma is a capacitively coupled plasma (CCP), and where the chamber part is a second electrode disposed in an upper portion of the plasma etch chamber.

[0065] Example 5. The method of one of examples 1 to 4, further including controlling a temperature of the second electrode to tune the degree of the sputtering from the second electrode.

[0066] Example 6. The method of one of examples 1 to 5, where the plasma etch chamber further including a focus ring, the focus ring surrounding the substrate and having a surface including another refractory metal, the method further including exposing the surface of the focus ring to the plasma to sputter the other refractory metal from the surface of the focus ring.

[0067] Example 7. The method of one of examples 1 to 6, further including controlling a temperature of the focus ring to tune the degree of the sputtering from the focus ring.

[0068] Example 8. The method of one of examples 1 to 7, further including applying a direct current (DC) voltage or a radio frequency (RF) power to the second electrode to tune the degree of the sputtering from the second electrode.

[0069] Example 9. The method of one of examples 1 to 8, where the deposited refractory metal on the portion of the patterned hard mask layer forms a metal carbide that preserves the portion of the patterned hard mask layer from being etched by the plasma.

[0070] Example 10. The method of one of examples 1 to 9, where the process gas includes fluorine and carbon.

[0071] Example 11. A plasma etching system for a substrate including: an etch chamber; a substrate holder disposed in the etch chamber; a top electrode disposed in the etch chamber, the top electrode having a surface including a refractory metal; a bottom electrode connected to the substrate holder; a first radio frequency (RF) power source connected to the bottom electrode, the first RF power source being configured to generate a plasma in the etch chamber and to sputter the refractory metal from the surface of the top electrode; and a focus ring disposed on the substrate holder and configured to surround the substrate.

[0072] Example 12. The plasma etching system of example 11, further including a second radio frequency (RF) power source connected to the bottom electrode, the second RF power source being configured to supply a bias to the plasma.

[0073] Example 13. The plasma etching system of one of examples 11 or 12, further including a direct current (DC) voltage source connected to the top electrode, the DC voltage source being configured to supply a DC voltage to the top electrode.

[0074] Example 14. The plasma etching system of one of examples 11 to 13, further including a third RF power source connected to the top electrode, the third RF power source being configured to supply a RF power to the top electrode.

[0075] Example 15. The plasma etching system of one of examples 11 to 14, further including a fourth radio frequency (RF) power source connected to the focus ring.

[0076] Example 16. The plasma etching system of one of examples 11 to 15, further including: a first sensor coupled to the top electrode to measure the temperature of the top electrode; a first heating element configured to heat the top electrode; and a first temperature controller coupled to the first sensor and the first heating element, the first temperature controller configured to control the heating element based on the measured temperature of the top electrode.

[0077] Example 17. The plasma etching system of one of examples 11 to 16, further including: a second sensor coupled to the bottom electrode or the focus ring to measure the temperature of the bottom electrode or the focus ring; a second heating element configured to heat the respective bottom electrode or the focus ring; and a second temperature controller coupled to the second sensor and the second heating element, the second temperature controller configured to control the second heating element based on the respective measured temperature of the bottom electrode or the focus ring.
[0078] Example 18. The plasma etching system of one of examples 11 to 17, where the focus ring has a top surface including another refractory metal.
[0079] Example 19. A plasma etching system for a substrate including: an etch chamber, where an upper wall inside the etch chamber including a refractory metal; a substrate holder disposed in the etch chamber; a top electrode including a helical coil disposed outside the etch chamber, the top electrode surrounding a upper portion of the etch chamber; a bottom electrode connected to the substrate holder; a radio frequency (RF) power source connected to the top electrode, the first RF power source being configured to generate a plasma in the etch chamber, the plasma being configured to sputter the metal from the upper wall; and a focus ring disposed on the substrate holder and configured to surround the substrate.
[0080] Example 20. The plasma etching system of example 19, where the focus ring includes another refractory metal.
[0081] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method for processing a substrate, the method comprising: loading the substrate into a plasma etch chamber, the substrate comprising a patterned hard mask layer and an underlying layer, the plasma etch chamber comprising: a chamber part having a surface comprising a refractory metal; and a first electrode; flowing a process gas into the plasma etch chamber; while flowing the process gas, applying a source power to the first electrode of the plasma etch chamber to generate a plasma in the plasma etch chamber; exposing the surface of the chamber part to the plasma to sputter the refractory metal from the surface of the chamber part; and exposing the substrate to the plasma to deposit the refractory metal onto a portion of the patterned hard mask layer and etch the underlying layer selectively to the patterned hard mask layer.
2. The method of claim 1, wherein the refractory metal is tungsten, molybdenum, niobium, tantalum, or ruthenium.
3. The method of claim 1, wherein the plasma is an inductively coupled plasma (ICP), and wherein the chamber part is a top plate disposed at an upper wall of the plasma etch chamber.
4. The method of claim 1, wherein the plasma is a capacitively coupled plasma (CCP), and wherein the chamber part is a second electrode disposed in an upper portion of the plasma etch chamber.
5. The method of claim 4, further comprising controlling a temperature of the second electrode to tune a degree of the sputtering from the second electrode.
6. The method of claim 4, wherein the plasma etch chamber further comprising a focus ring, the focus ring surrounding the substrate and having a surface comprising an second refractory metal, the method further comprising exposing the surface of the focus ring to the plasma to sputter the second refractory metal from the surface of the focus ring.
7. The method of claim 6, further comprising controlling a temperature of the focus ring to tune a degree of the sputtering from the focus ring.

8. The method of claim 4, further comprising applying a direct current (DC) voltage or a radio frequency (RF) power to the second electrode to tune a degree of the sputtering from the second electrode.

9. The method of claim 1, wherein the deposited refractory metal on the portion of the patterned hard mask layer forms a metal carbide that preserves the portion of the patterned hard mask layer from being etched by the plasma.

10. The method of claim 1, wherein the process gas comprises fluorine and carbon.

11. A method for processing a substrate, the method comprising: processing a gas that flows through a chamber of an etch chamber, the etch chamber further comprising a first electrode and a substrate holder to hold a substrate, the substrate comprising a patterned hard mask layer and an underlying layer, the chamber having a first surface comprising a first refractory metal, the substrate holder having a focus ring surrounding the substrate and having a top surface coated with a second refractory metal for metal sputtering, the first refractory metal and the second refractory metal being different refractory metals; and generating a plasma in the etch chamber by applying an RF power from an RF power source to the first electrode during the processing of the gas flowing through the chamber, the RF power source coupled to the first electrode, the plasma exposed to the first surface to sputter the first refractory metal and to the substrate to deposit the first refractory metal and the second refractory metal onto a portion of the patterned hard mask layer and to etch the underlying layer selectively to the patterned hard mask layer.

12. The method of claim 11, wherein each of the first refractory metal and the second refractory metal are a same refractory metal is independently selected from the group consisting of tungsten, molybdenum, niobium, tantalum, and ruthenium.

13. The method of claim 11, wherein the plasma is an inductively coupled plasma (ICP), and wherein the etch chamber further includes a top plate disposed at an upper wall of the etch chamber.

14. The method of claim 11, wherein the plasma is a capacitively coupled plasma (CCP), and wherein the etch chamber includes a second electrode disposed in an upper portion of the etch chamber.

15. The method of claim 14, further comprising controlling, by a temperature controller, a temperature of the second electrode to tune a degree of the sputtering from the second electrode.

16. The method of claim 14, further comprising applying a DC voltage to the second electrode to tune a degree of the sputtering from the second electrode or a second RF source, the second RF source used to apply a second RF power to the second electrode to tune a degree of the sputtering from the second electrode.

17. The method of claim 11, further comprising a controlling, by a temperature controller, a temperature of the focus ring to tune a degree of the sputtering from the second refractory metal.

18. The method of claim 11, wherein the first refractory metal and the second refractory metal deposited on the portion of the patterned hard mask layer form a metal carbide that preserves the portion of the patterned hard mask layer from being etched by the plasma.

19. A method for processing a substrate in an etch chamber, the method comprising: generating, by a first radio frequency (RF) power source of the etch chamber, plasma in a chamber of the etch chamber comprising a substrate holder to hold a substrate and having a focus ring surrounding the substrate, a top electrode having a first refractory metal, and a bottom electrode coupled to the substrate holder and the first RF power source, the focus ring having a top surface coated with a second refractory metal different from the first refractory metal, wherein the plasma causes the first refractory metal to sputter from a surface of the top electrode, wherein the plasma causes the second refractory metal to be a second source of refractory metal for metal sputtering.

20. The method of claim 19, wherein the etch chamber further comprises a second RF power source coupled to the bottom electrode, a direct current (DC) voltage source and a third RF power source coupled to the top electrode, the method further comprising: supplying, by the second RF power source, a bias to the plasma; supplying, by the DC voltage source, a DC voltage to the top electrode; and supplying, by the third RF power source, a second RF power to the top electrode.